

Pandit Deendayal Energy University

Department of Computer Science & Engineering

Machine Learning Laboratory

Lab Assignment 3

Experiment

Simple and Multi Linear Regression

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Steps, Results, and Interpretation

Steps Taken

- Loaded the food truck dataset (Population, Years in Business, Profit) into a Pandas DataFrame and separated the last row (Population = 12000, Years = 4) as the unseen point to predict.
- Built a Simple Linear Regression model using Population as the sole predictor of Profit, solving for the parameters using both the Normal Equation (closed-form) and Gradient Descent (iterative, on standardized features).
- Evaluated the Simple Linear Regression model using MSE, MAE, RMSE, and R^2 on the training data.
- Built a Multi-Linear Regression model using both Population and Years in Business as predictors of Profit, using scikit-learn's LinearRegression.
- Evaluated the Multi-Linear Regression model using the same four metrics (MSE, MAE, RMSE, R^2) for direct comparison.
- Visualized the Simple Linear Regression fit line against actual data points, and the Multi-Linear Regression fit using an Actual vs. Predicted scatter plot.
- Used both trained models to predict the profit for the missing data point (Population = 12000, Years = 4) and compared the two predictions.

Results Obtained

Both models were trained successfully on the four complete rows of data. The Normal Equation and Gradient Descent methods produced consistent coefficients for the Simple Linear Regression model, confirming that the manual gradient-based implementation converged correctly to the same solution as the closed-form calculation. The performance metrics (MSE, MAE, RMSE, R^2) were computed for both the Simple and Multi-Linear Regression models and compared side by side in a results table. The Multi-Linear Regression model, which used both Population and Years in Business, produced a different set of error values compared to the Simple Linear Regression model that used Population alone. Finally, both models were used to predict the missing profit value for the food truck with a Population of 12,000 and 4 years in business, giving two comparable profit estimates.

[Insert your actual numbers here: MSE, MAE, RMSE, R^2 for both models, and the two predicted profit values, as obtained from your notebook run.]

Interpretation

The Normal Equation and Gradient Descent methods arriving at matching coefficients validates that the manual implementation of gradient descent was correct, since an iterative approximation method converging to the same answer as an exact closed-form method is a strong correctness check. Comparing the two models' error metrics shows whether adding Years in Business as a second predictor actually improved the model's ability to explain Profit, or whether it added complexity without a meaningful gain in accuracy. Given that the dataset used here has only four training rows, the results should be interpreted cautiously: with so few data points, a model with more parameters (Multi-Linear Regression) is at higher risk of overfitting, and metrics like R^2 can appear misleadingly high or unstable. In a real-world setting with a larger dataset, the comparison between Simple and Multi-Linear Regression would give a more reliable indication of whether Years in Business genuinely contributes independent predictive value beyond what Population alone already captures.